Differential abundance analysis

0.1 Differential abundance analysis

Analysis information:

- Taxonomic level: species_prevalent

Taxa that are less prevalent than this (in relabundance) have been agglomerated:

- Detection threshold: 0.1%
- Prevalence threshold: 0.0714286
- Number of taxonomic groups: 215

BMI is not significantly different between conditions or time points. Hence we can ignore it in the differential abundance testing. In group "B" it seems to be borderline significant ($p<0.05$ before multiple testing adjustment).

Number of significant findings within time points.

- Kruskal test, Benjamini-Hochberg FDR adjustment

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before  after
     1     14
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Number of significant findings within treatments:

- Paired Wilcoxon test, Benjamini-Hochberg FDR adjustment

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rice  oat
  14    8
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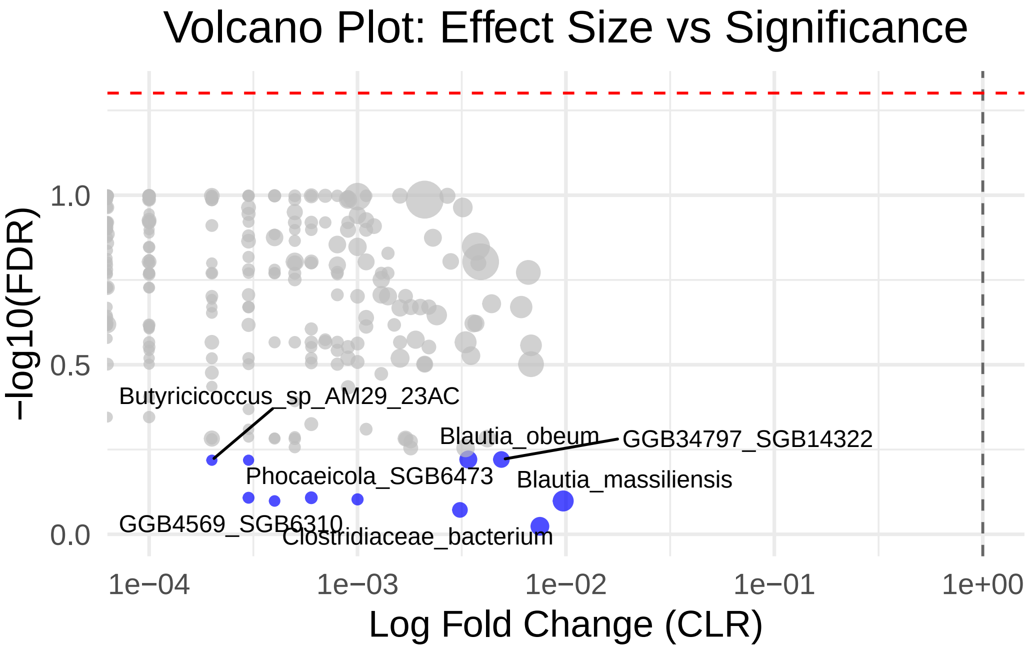
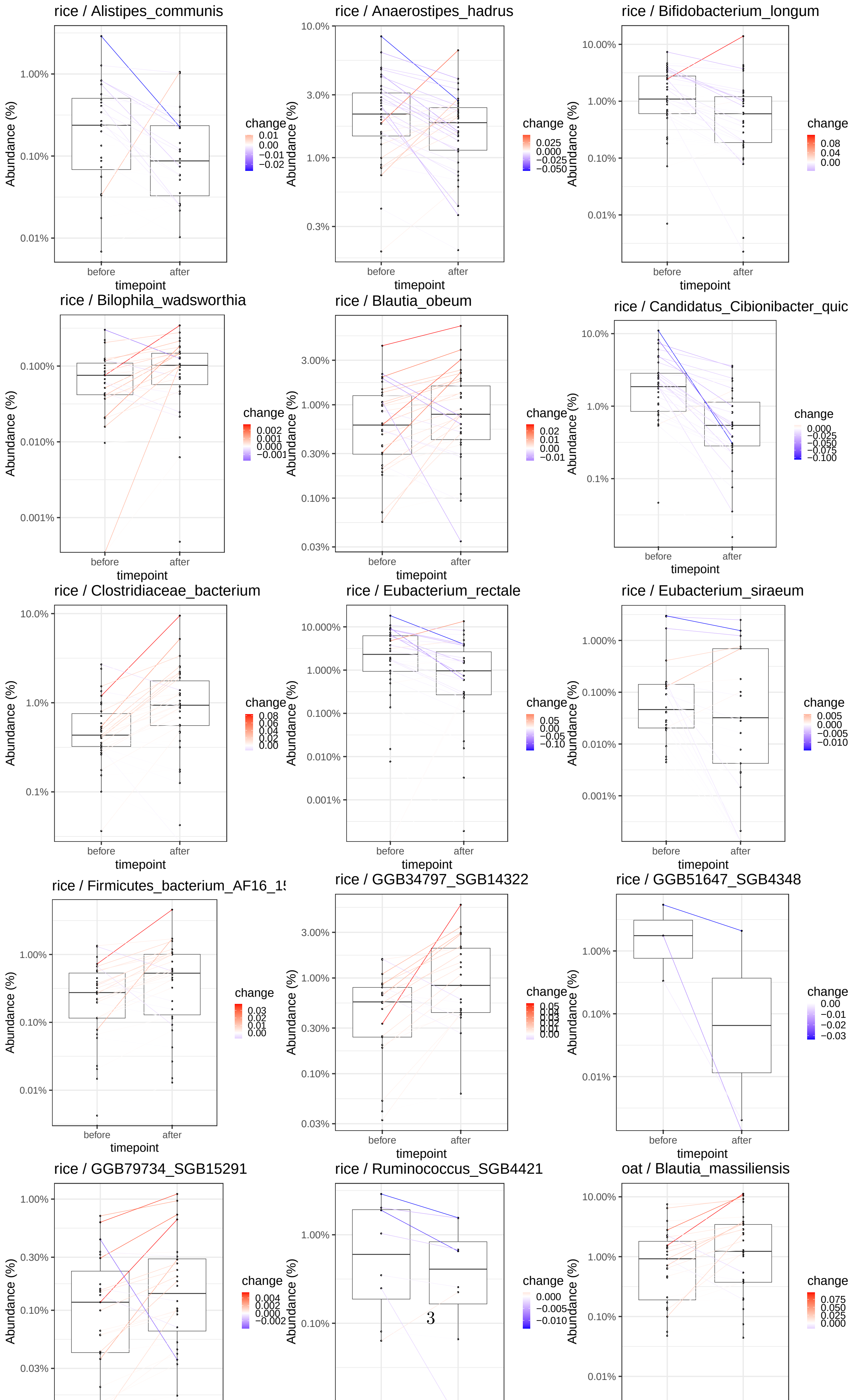
Let us look more closely to the significant hits. The logFC corresponds to logFC in CLR (tp2-tp1).

Group	Taxon	logFC.clr	FDR	mean.relabundance1	mean.relabundance2
rice	Candidatus_Cibionibacter_quicibialis	-0.0147	0.0003	2.336	0.822
rice	Bifidobacterium_longum	-0.0049	0.0019	1.794	1.311
rice	Clostridiaceae_bacterium	0.0075	0.0236	0.648	1.422
rice	Firmicutes_bacterium_AF16_15	0.0031	0.0719	0.317	0.627
oat	Blautia_massiliensis	0.0097	0.0983	1.169	2.183
oat	GGB4569_SGB6310	0.0004	0.0983	0.029	0.070
oat	Candidatus_Cibionibacter_quicibialis	-0.0052	0.0983	1.548	1.014
oat	GGB27106_SGB6188	0.0010	0.1030	0.019	0.115
rice	GGB79734_SGB15291	0.0006	0.1078	0.116	0.178
rice	Bilophila_wadsworthia	0.0003	0.1078	0.066	0.096
rice	Eubacterium_rectale	-0.0127	0.1238	3.214	1.878
oat	Clostridiales_bacterium_KLE1615	-0.0025	0.1552	0.667	0.412
rice	Eubacterium_siraeum	-0.0006	0.1991	0.262	0.205
oat	Lawsonibacter_asaccharolyticus	-0.0005	0.2078	0.164	0.111
oat	Phocaeicola_SGB6473	0.0003	0.2185	0.023	0.048
oat	Butyricicoccus_sp_AM29_23AC	0.0002	0.2185	0.033	0.049
rice	GGB34797_SGB14322	0.0049	0.2206	0.355	0.851
rice	Blautia_obeum	0.0034	0.2206	0.832	1.179
rice	Ruminococcus_SGB4421	-0.0010	0.2206	0.245	0.142
rice	GGB51647_SGB4348	-0.0015	0.2206	0.215	0.060

Group	Taxon	logFC.clr	FDR	mean.relabundance1	mean.relabundance2
rice	Alistipes_communis	-0.0016	0.2206	0.295	0.135
rice	Anaerostipes_hadrus	-0.0055	0.2206	2.495	1.923




rice / Alistipes_communis
rice / Anaerostipes_hadrus
rice / Bifidobacterium_longum
rice / Bilophila_wadsworthia
rice / Blautia_obeum
rice / Candidatus_Cibionibacter_quic
rice / Clostridiaceae_bacterium
rice / Eubacterium_rectale
rice / Eubacterium_siraeum
rice / Firmicutes_bacterium_AF16_1!
rice / GGB34797_SGB14322
rice / GGB51647_SGB4348
rice / GGB79734_SGB15291
rice / Ruminococcus_SGB4421
oat / Blautia_massiliensis
Abundance (%)
timepoint
before
after
change